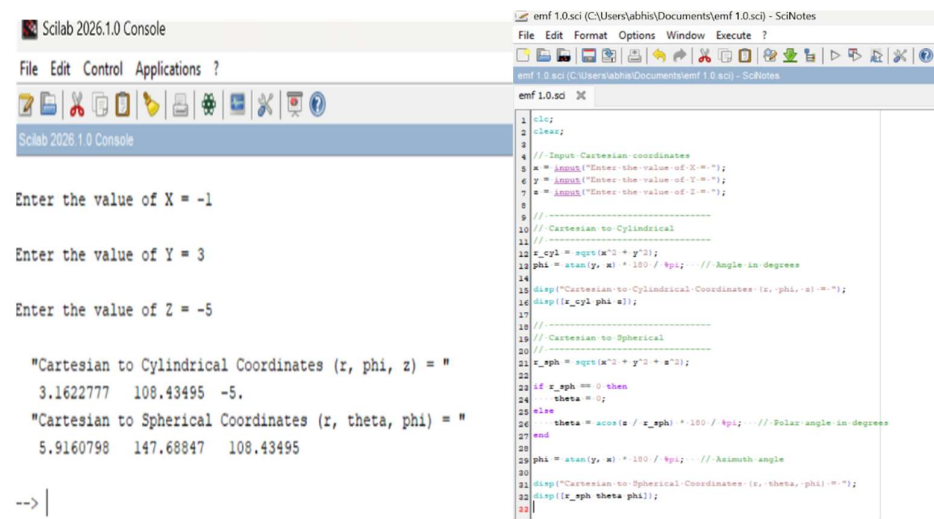


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Emf_eca0502

Aim:

To write program for Transformation between Cartesian, Cylindrical and spherical coordinate system using SCILAB.



```
Scilab 2026.1.0 Console
File Edit Control Applications ?
Scilab 2026.1.0 Console

Enter the value of X = -1

Enter the value of Y = 3

Enter the value of Z = -5

"Cartesian to Cylindrical Coordinates (r, phi, z) = "
3.1622777 108.43495 -5.
"Cartesian to Spherical Coordinates (r, theta, phi) = "
5.9160798 147.68847 108.43495

-->
```

```
emf 1.0.sci (C:\Users\abhis\Documents\emf 1.0.sci) - SciNotes
File Edit Format Options Window Execute ?
emf 1.0.sci (C:\Users\abhis\Documents\emf 1.0.sci) - SciNotes
emf 1.0.sci 36

1 clear;
2 clear;
3
4 // Input Cartesian coordinates
5 x = input("Enter the value of X :- ");
6 y = input("Enter the value of Y :- ");
7 z = input("Enter the value of Z :- ");
8
9 // -----
10 // Cartesian to Cylindrical
11 // -----
12 r_cyl = sqrt(x^2 + y^2);
13 phi = atan(y, x) * 180 / %pi; // Angle in degrees
14
15 disp("Cartesian to Cylindrical Coordinates (r, phi, z) :- ");
16 disp([r_cyl phi z]);
17
18 // -----
19 // Cartesian to Spherical
20 // -----
21 r_sph = sqrt(x^2 + y^2 + z^2);
22
23 if r_sph == 0 then
24     theta = 0;
25 else
26     theta = acos(z / r_sph) * 180 / %pi; // Polar angle in degrees
27 end
28
29 phi = atan(y, x) * 180 / %pi; // Azimuth angle
30
31 disp("Cartesian to Spherical Coordinates (r, theta, phi) :- ");
32 disp([r_sph theta phi]);
33
```

Exp-1

for the given values:

$$x = -1$$

$$y = 3$$

$$z = -5$$

Formula:

$$r = \sqrt{x^2 + y^2}$$

$$\phi = \arctan_2(y, x)$$

$$z = z$$

Calculation:

$$r = \sqrt{(-1)^2 + 3^2} = \sqrt{10} = 3.1623$$

$$\phi = \arctan_2(3, -1) = 108.435^\circ$$

$$z = -5$$

$$(r, \phi, z) = (3.1623, 108.435^\circ, -5)$$

Cartesian to Spherical Coordinates:

$$R = \sqrt{x^2 + y^2 + z^2}$$

$$\theta = \cos^{-1}\left(\frac{z}{R}\right)$$

$$\phi = \arctan_2(y, x)$$

Calculation:

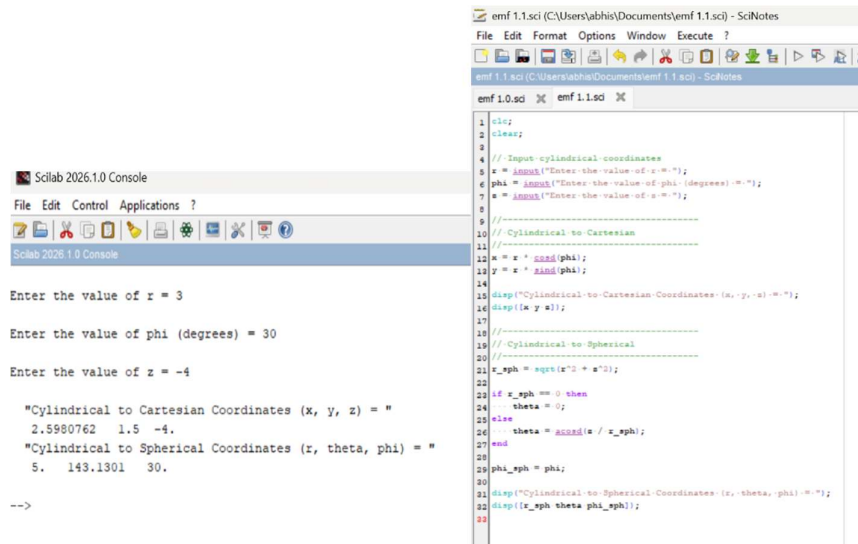
$$R = \sqrt{(-1)^2 + 3^2 + (-5)^2} = \sqrt{35} = 5.9161$$

$$\theta = \cos^{-1}\left(\frac{-5}{5.9161}\right) = 147.688^\circ$$

$$\phi = 108.435^\circ$$

$$(R, \theta, \phi) = (5.9161, 147.688^\circ, 108.435^\circ)$$

Test case (ii) Transformation of Cylindrical coordinate points into Cartesian and spherical coordinate system



The image shows two windows from the Scilab environment. The left window is the 'Scilab 2026.1.0 Console', which displays the results of a script execution. The right window is a script editor showing the source code of 'emf 1.1.sci'.

Scilab 2026.1.0 Console Output:

```
Enter the value of r = 3

Enter the value of phi (degrees) = 30

Enter the value of z = -4

"Cylindrical to Cartesian Coordinates (x, y, z) ="
2.5980762  1.5  -4.

"Cylindrical to Spherical Coordinates (r, theta, phi) ="
5.  143.1301  30.

-->
```

emf 1.1.sci Script Code:

```
1 clear;
2 close;
3
4 // Input cylindrical coordinates
5 r = input("Enter the value of r = ");
6 phi = input("Enter the value of phi (degrees) = ");
7 z = input("Enter the value of z = ");
8
9 //-----
10 // Cylindrical to Cartesian
11 //-----
12 x = r * cosd(phi);
13 y = r * sind(phi);
14
15 disp("Cylindrical to Cartesian Coordinates (x, y, z) = ");
16 disp([x y z]);
17
18 //-----
19 // Cylindrical to Spherical
20 //-----
21 r_sph = sqrt(x^2 + y^2);
22
23 if r_sph == 0 then
24     theta = 0;
25 else
26     theta = asind(x / r_sph);
27 end
28
29 phi_sph = phi;
30
31 disp("Cylindrical to Spherical Coordinates (r, theta, phi) = ");
32 disp([r_sph theta phi_sph]);
33
```

Given Values:

$$\begin{aligned}r &= 3 \\ \phi &= 30^\circ \\ z &= -4\end{aligned}$$

Formula:

$$\begin{aligned}x &= r \cos \phi \\ y &= r \sin \phi \\ z &= z\end{aligned}$$

Calculation:

$$x = 3 \times \cos(30^\circ)$$

$$x = 3 \times 0.866 = 2.5981$$

$$y = 3 \times \sin(30^\circ)$$

$$y = 3 \times 0.5 = 1.5000$$

$$z = -4$$

$$(x, y, z) = (2.5981, 1.5000, -4)$$

Formula:

$$R = \sqrt{x^2 + y^2 + z^2}$$

$$\theta = \cos^{-1}\left(\frac{z}{R}\right)$$

$$\phi = \phi$$

Calculation:

$$R = \sqrt{3^2 + (-4)^2} = \sqrt{9+16} = \sqrt{25} = 5$$

$$\theta = \cos^{-1}\left(\frac{-4}{5}\right) = \cos^{-1}(-0.8) = 143.13^\circ$$

$$\phi = 30^\circ$$

$$(R, \phi, \theta) = (5, 30^\circ, 143.13^\circ)$$

Test case (iii) Transformation of spherical coordinate points into Cylindrical and Cartesian coordinate system

```

1  clc;
2  clear;
3
4  // Input spherical coordinates
5  r = input("Enter the value of r = ");
6  phi = input("Enter the value of phi (degrees) = ");
7  theta = input("Enter the value of theta (degrees) = ");
8
9  // Spherical to Cartesian
10 //-----
11
12 x = r * sind(theta) * cos(phi);
13 y = r * sind(theta) * sind(phi);
14 z = r * cos(theta);
15
16 disp("Spherical to Cartesian Coordinates (x, y, z) = ");
17 disp([x y z]);
18
19 // Spherical to Cylindrical
20 //-----
21
22 r_cyl = r * sind(theta);
23 phi_cyl = phi;
24 z_cyl = r * cos(theta);
25
26 disp("Spherical to Cylindrical Coordinates (r, phi, z) = ");
27 disp([r_cyl phi_cyl z_cyl]);
28

```

Scilab 2026.1.0 Console

```

File Edit Control Applications ?
Scilab 2026.1.0 Console

Enter the value of r = 10
Enter the value of phi (degrees) = 45
Enter the value of theta (degrees) = 130

"Spherical to Cartesian Coordinates (x, y, z) = "
5.4167522 5.4167522 -6.4278761
"Spherical to Cylindrical Coordinates (r, phi, z) = "
7.6604444 45. -6.4278761
--> |

```

$r = 10$
 $\phi = 45^\circ$
 $\theta = 130^\circ$

$x = r \sin \theta \cos \phi$
 $y = r \sin \theta \sin \phi$
 $z = r \cos \theta$

$\sin 130^\circ = 0.7660444$
 $\cos 130^\circ = -0.6427876$
 $\sin 45^\circ = 0.7071068$
 $\cos 45^\circ = 0.7071068$

$x = 10 \times 0.7660444 \times 0.7071068$
 $x = 5.4167522$
 $y = 10 \times 0.7660444 \times 0.7071068$
 $y = 5.4167522$
 $z = 10 \times (-0.6427876)$
 $z = -6.4278761$

$P(X, Y, Z) = (5.4167522, 5.4167522, -6.4278761)$
Formula
 $\rho = r \sin \theta$
 $\phi = \phi$
 $z = r \cos \theta$
 $\rho = 10 \times 0.7660444$
 $\rho = 7.660444$
 $z = 10 \times (-0.642787876)$
 $z = -6.4278781$
 $P(\rho, \phi, z) = (7.6604444, 45^\circ, -6.4278761)$
 $P(X, Y, Z) = (5.4167522, 5.4167522, -6.4278761)$
 $P(\rho, \phi, z) = (7.6604444, 45^\circ, -6.4278761)$

Result:

Thus, the program was verified Transformation between Cartesian and Cylindrical coordinate system its

classified two categories (a) Cartesian to cylindrical and spherical coordinate system (b) Cylindrical to

cartesian and spherical coordinate system (c) spherical to cartesian and cylindrical coordinate system

using SCILAB was successfully.